

1 • Input

Molecular formula + literature
peak lists (IR cm^{-1} , ^1H , ^{13}C)

blind peak lists (not traces)

2 • Generation

LLM proposes ranked
candidate structures

true enters top-3 pool 249/500 (49.8%)

3 • Verification

Forward-verify / re-rank
candidates against spectra

among recalled, top-1 227/249 (91.2%)

top-1 = recall@3 × (top-1 | recalled)

45.4% = 49.8% × 91.2%

top-1 227/500 = 45.4%; recall@3 249/500 = 49.8%; among recalled 227/249 = 91.2%
249 recalled (49.8%)

227

22

251

exact top-1

in pool, not top-1

never proposed

No re-ranking repairs the 251 never proposed in top-3